

DHARANESH PALANIAPPAN

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PROFILE

Highly proficient in computational and observational astrophysics. Proven ability to apply specialized tools to research, evidenced by projects and publications. My core research interests lie in Exoplanetary Science, Helio Physics, and Planetary Science.

EDUCATION

- Master of Science in Physics** | June 2025 - Present
Madras Christian College
- Bachelor of Science in Physics** – Final grade 8.06/10 | July 2022 – April 2025
Madras Christian College

RESEARCH EXPERIENCE

- Comprehensive Analysis of 501 Stars and Their Exoplanetary Systems Using Lightcurve and Exoplanet Data** (Present)
Conducting analysis on 500 stars selected from the MAST database to search for exoplanet signals and crossmatch with confirmed hosts. Using light-curve analysis, HR diagram visualization, and statistical correlations to study stellar properties that favor planet hosting.
- Solar Remote Sensing and Active Region Studies** (Present)
Using remote sensing data from NASA's SDO instruments (AIA & HMI) and data from Parker Solar Probe (PSP) to study solar active regions and solar activities. Under guidance of Mr. Khagendra Katuwal | PhD candidate, New Mexico State University

PUBLICATIONS

- Preliminary analysis of a potential asteroid from the IASC campaign using Pan-STARRS data** (Preprint)
Published a paper presenting the preliminary analysis of a potential asteroid, with its identification confirmed by the Minor Planet Centre. Employed astrometric and photometric techniques using data from the International Astronomical Search Collaboration (IASC) and Pan-STARRS.

ROLES & EXPERIENCE

- B.M. ARTHUR ASTRO-EPHEMERIS CLUB SECRETARY** (Madras Christian College) – July 2022 – April 2025
Organized and led the astronomy activities, including telescope-based stargazing sessions and special celestial-event programs, handling telescope setup, alignment, and operation using Celestron systems and software.
- INTERNATIONAL ASTEROID SEARCH CAMPAIGN** – Feb 2025 – Present
Analyzed astronomical data using Astrometrica to identify asteroids in a NASA-affiliated citizen science campaign. Reported findings for official verification, gaining practical experience in astrometry, data analysis, and data reduction.

SKILLS & KNOWLEDGE

Python, Lightcurve, SunPy, UNIX, Astrometrica, Celestron software, Telescope Handling, Electronics

SUMMER SCHOOL & WORKSHOPS

- **Intro2Astro – Sep 2025**
Participated in the Intro2Astro summer workshop, which is designed for undergraduate students. In the summer workshop, we learned the basics of Python, data analysis with real astronomical data, and future career opportunities.
- **The South Asian Astrophysics Summer School 2025 – July 2025**
Analyzed real-time solar data using both remote and in situ measurement techniques. This included working with data from the Parker Solar Probe, HMI, and AIA instruments, which enhanced my practical skills in solar observation and advanced data analysis.
- **ZTF Summer School – July 2025**
Participated in advanced training on time-domain astronomy with the Zwicky Transient Facility, focusing on light curve analysis, transient detection, and large-scale survey data processing using Python-based tools.
- **Sagan Summer Workshop 2025 – July 2025**
Engaged in specialized training on exoplanet detection and characterization techniques, with emphasis on light curve analysis, transit methods, and statistical tools for exoplanetary science.

CONFERENCES AND SEMINARS

- **10 years of Gravitational Wave Astronomy** | Indian Institute of Technology, Madras – Sep 2025
- **Youth Astronomy and Space Science Congress**, Virudhunagar - July 2025
- **Online review writing series** | Madras Christian College - Mar 2025
- **CERN 70th anniversary** | Indian Institute of Technology, Madras – Feb 2025
- **High Energy Materials** | Srihari Kota (SDSC SHAR) – Sep 2024
- **Asteroid Bennu: A Glimpse into What Lies Beyond** | Loyola College, Remote – Dec 2023
- **Remembering Amal Kumar Raychaudhuri (AKR)** | Institute of Mathematical Sciences, Chennai – Oct 2023
- **Prospects of Experiential learning in Material Physics** | Manipal University, Remote – Jul 2023

CERTIFICATIONS & COURSES

- International Astronomy and Astrophysics Competition - Silver Honor
- Confronting The Big Questions: Highlights of Modern Astronomy University of Rochester (Via Coursera).
- Astronomy: Exploring Time and Space University of Arizona (Via Coursera)
- The Science of The Solar System Caltech (Via Coursera)
- The Diversity of Exoplanets University of Geneva (Via Coursera)

VOLUNTEERING

- **Young Student Scientist Programme** (Jan–Feb 2024) Mentored students in interactive science sessions and demonstrations.
- **SCIIGNITE: Fanning the Flames of Science Passion** (Jan 2024) Guided Students in basic electronics and hands-on experiments.
- **Star Party** (Jan 2023) Assisted with telescope handling, night-sky navigation, and astronomy outreach.
- **Devadas Observatory** (2023- 2024) Assisted the Observatory on telescope handling.